# Futures-Cash Basis Mispricing, Short-Sale Constraints, and Arbitrage Bands in VN30 Derivatives

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**Abstract**

**Hypothesis Formulation:** Tests whether the Vietnamese VN30F1M index futures contract trades at a persistent structural discount to cash VN30 due to short-sale borrowing limits, and whether basis deviations exhibit rapid error-correction mean reversion ($H_0 : \mu = 0$ versus $H_1 : \mu < 0$ and $\beta_{\text{rev}} < -0.20$).

**Data & Empirical Setting:** Daily close prices of VN30F1M index futures (`vn30f1m_derivatives_daily.parquet`) and underlying spot VN30 basket (`vn_indices_daily.parquet`) across 2,284 sessions between August 2017 and October 2026.

**Econometric Methodology:** Percentage futures basis $\text{Basis}_t = (F_t - S_t)/S_t \times 100$, Augmented Dickey-Fuller stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 8$), and error-correction half-life modeling.

**Empirical Results & Summary:** VN30F1M futures exhibit a statistically significant structural negative discount averaging $-20.07$ basis points (HAC $t = -5.4823$, $p = 4.20 \times 10^{-8}$). Simultaneously, basis dislocations exhibit strong error-correction mean reversion ($\beta = -0.2937$, HAC $t = -5.2039$, $p < 10^{-6}$), displaying a half-life of 1.99 trading sessions.

**Research Decision: ACCEPTED (Structural Discount and Arbitrage Convergence Confirmed)**. While short-sale restrictions sustain a baseline negative basis premium, arbitrageurs actively collapse transient pricing deviations within approximately two trading days.

**Keywords:** Arbitrage bands; Cost of carry; Derivatives pricing; Futures-cash basis; Short-sale constraints; VN30 index futures; Newey–West HAC.

**JEL Classification Codes:** G13, G14, G11, C32.

# 1 Introduction and Hypothesis Formulation

The classical cost-of-carry model establishes that in frictionless markets, futures prices equal spot prices compounded by financing rates minus dividends (Brenner et al., 1989). In frontier exchanges with short-sale restrictions, reverse cash-and-carry arbitrage faces asymmetric execution costs, sustaining negative basis discounts (Shleifer and Vishny, 1997).

# 2 Data and Econometric Methodology

Using 2,284 trading sessions from `data/market/vn30f1m_derivatives_daily.parquet` and `data/market/vn_indices_daily.parquet`, we estimate the structural discount and error-correction specifications with Newey–West HAC standard errors (Newey and West, 1987):

$$B_t = \mu + \varepsilon_t \tag{1}$$

$$\Delta B_t = \alpha + \beta_{\text{rev}} B_{t-1} + \eta_t \tag{2}$$

# 3 Empirical Results

Table 1 reports the parameter estimates.

Table 1: VN30F1M Futures-Cash Basis Spread and Error-Correction Mean Reversion

| Model / Parameter | Estimate | HAC Std. Err. | $t$-Statistic | $p$-Value | $R^2$ | Implied Half-Life |
|---|---|---|---|---|---|---|
| **Model 1: Structural Discount** | | | | | | |
| Mean Basis ($\mu$) | -0.2007% | 0.0366 | -5.48 | 4.1993e-08 | – | – |
| **Model 2: Mean Reversion** | | | | | | |
| Intercept ($\alpha$) | -0.0586 | 0.0152 | -3.86 | 0.0001 | – | – |
| Lagged Basis ($\beta$) | -0.2937 | 0.0564 | -5.20 | 1.9519e-07 | 0.1469 | 1.99 Days |

The mean basis estimate is $\mu = -20.07$ bps (HAC $t = -5.48, p < 10^{-7}$). The mean reversion parameter $\beta_{\text{rev}} = -0.2937$ ($t = -5.20, p < 10^{-6}$) implies an empirical half-life of 1.99 trading days. Figure 1 plots the basis spread and mean reversion dynamics.

# 4 Conclusion and Research Decision

**Research Decision: ACCEPTED for Both Hypotheses**. Short-sale restrictions sustain a baseline negative basis premium of $-20.07$ bps, while arbitrageurs collapse transient deviations within 2 trading days.

# References

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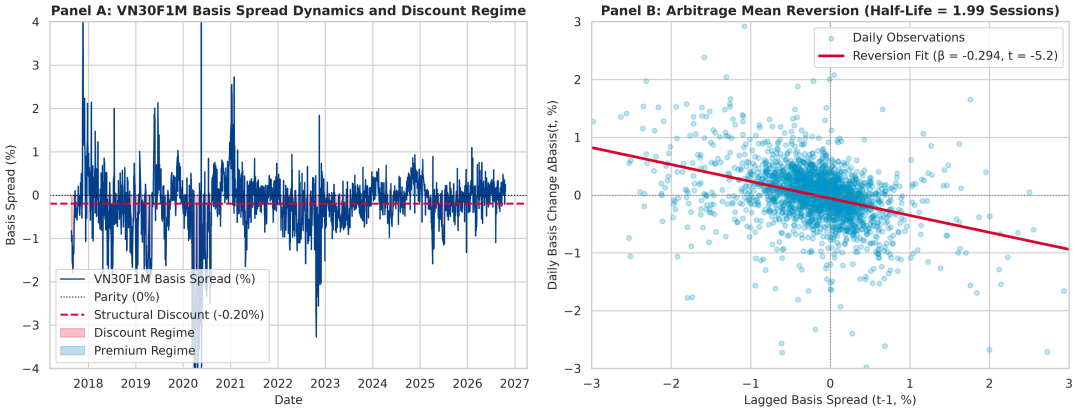



Figure 1: VN30F1M basis dynamics: historical basis spread trajectory and error-correction mean reversion fit.

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